

Development of a Computational Algorithm for the Automation of Quality Control in Leaded Personal Protective Equipment

Jerónimo Gil Castañeda^a and Erika Muñoz Arango^b

^a Universidad Nacional de Colombia, Facultad de Ciencias Exactas y Naturales, Manizales, Colombia.

^b Área de Física Médica Hospitalaria, Servicios Especiales de Salud, Hospital Universitario de Caldas (SES-HUC), Manizales, Colombia.

E-mail: jeronimogil91@gmail.com

ABSTRACT

An image processing algorithm was developed using Python to automate the quality control of leaded personal protective equipment (PPE-Pb) employed by occupationally exposed personnel. Digital images of these protective garments were acquired via computed tomography and subsequently converted from DICOM to grayscale PNG format to facilitate crack detection and area quantification.

The algorithm demonstrated accurate classification capabilities for vests and thyroid shields, and successfully identified cracks in lead aprons. The measured areas of the detected cracks ranged from 6.1 cm² to 12.5 cm². One apron was deemed non-compliant and rejected due to exceeding the permissible damage threshold.

These findings support the viability of the proposed algorithm as a reliable tool for the detection and quantification of structural damage in PPE-Pb. Its implementation has the potential to enhance radiological protection protocols by streamlining quality control procedures and minimizing observer dependency.

Keywords: PPE-Pb, Quality Control, Cracks, Image Processing, Rejection Criteria.

1. Introduction

Various national and international protocols and recommendations related to radiological protection recommend the use of leaded personal protective equipment (PPE-Pb) for the attenuation of scattered radiation generated by different medical procedures in radiotherapy and radiodiagnosis [1]–[6]. Documents from the International Commission on Radiological Protection (ICRP) [2]; ICRP [3]; U.S. Environmental Protection Agency (EPA) [5], and Ministerio de Minas y Energía [6], among others, recommend the constant use of leaded personal protective equipment (PPE-Pb), such as vests, thyroid protectors, and gonadal protectors in any situation where an occupationally exposed worker (OEW), patient, or member of the public is in an environment with ionizing radiation [1]–[5], [7].

Personal shielding is one of the basic radiological protection measures and is based on keeping the doses received from scattered radiation as low as possible. To achieve this, periodic quality controls of these elements are recommended in order to ensure the good condition of the internal material they are made of; among these, the material responsible for attenuating radiation is usually lead or an equivalent material [1], [2], [6], [7].

Cracks, fissures, and gaps that may appear in PPE-Pb as a result of use are indicators of material fatigue, reducing their lifespan and main function. The presence and total area of these cracks become a parameter to establish quantitative rejection criteria and subsequent replacement of PPE-Pb. Rejection criteria can have a cost/benefit optimization approach, considering the incremental doses to OEWs due to the use or disuse of PPE-Pb and the cost of replacing them; in this way, a maximum crack area is established for each PPE-Pb [1], [7]. Defects in the lead structure of these elements are detected through radiological imaging (conventional Xray plates or tomography) in order to obtain a digital image that allows visualization of their internal condition; this facilitates the application of evaluation criteria [1], [7].

Digital images from computed tomography (CT) are in DICOM format (Digital Imaging and Communication in Medicine), which contains visual information in Hounsfield Units (HU), obtained from a linear transformation of X-ray attenuation coefficients into a new scale. This transformation results in a scale ranging from -1000 HU for air to ± 2000 HU for very dense bones, and over +3000 HU for metals [8], [9], [10].

The DICOM format presents a two-part structure: the header and the dataset [9], [11], [12], [13]. The header contains information related to the encoding syntax and UID (user ID) of the file and the type of compression of the dataset [10], [14]. The dataset represents the object of study and consists of various data elements that encode the attributes and characteristics of the object, which are described by four types of tags: TAG, VR, VL, VF. The TAG is the unique identification tag, VR indicates the type or class of data stored (integer, real, etc.); VL denotes the length of the data (in bytes); VF contains the study information, such as patient data, equipment details, and radiological parameters used in the procedure [12].

Advancements in crack and fissure detection through image processing have allowed the application of different models in the industry; fissures in concrete and composite materials are an example of these applications [11], [14], [15], [16]. Various processing methods exist, including: mathematical morphology techniques, thresholding, spatial domain smoothing techniques, filters and cropping masks, edge and object detection, among others. Mathematical morphology is a strategy for analyzing certain special structures in an image using set theory, integral geometry, and lattice algebra. This technique uses elements such as the convolution matrix Kernel; structural elements to perform erosion, dilation, skeletonization, and others [17], [18].

Filtering and smoothing techniques in the spatial domain are based on direct pixel manipulation through an $M \times N$ neighborhood matrix (number of rows by number of columns), also known as the convolution Kernel matrix; that is, each pixel acquires properties based on the values of neighboring pixels in order to eliminate undesired noise [17], [18]. Adaptive filters are more complex as the weighting is calculated for each pixel based on its function [17], [19]. Thresholding involves separating the image into regions based on their intensity values using a threshold T . Objects with intensity values less than T belong to one region, while those with intensity greater than T belong to another, resulting in a two-level image or mask [17]–[19]. Edge detection is a technique for clipping objects within an image, based on detecting abrupt intensity changes in pixels using first and second derivatives [16], [17], [19].

In this work, a computational tool was developed to optimize the quality control of PPE-Pb, considering the acceptance or rejection criteria established by Phillips and Duran [1]. For this, a model based on digital image processing was followed for the morphological extraction of cracks present in lead, using a combination of adaptive filters and segmentation processes to ensure the internal condition of PPE-Pb, which aims to provide radiological protection.

2. Materials and methods

2.1. Image Acquisition of PPE-Pb

Using a Siemens SOMATOM go.Sim CT scanner, DICOM images of PPE-Pb were acquired through a topogram. A "Trauma Body [Factory]" protocol (110 kV - 20 mA) was followed, positioning each PPE-Pb from various service areas at the isocenter defined by the scanner's internal lasers.

2.2. Algorithm Development

For crack detection, it was necessary to transform the DICOM image. This was achieved using the Python 3.12.6 virtual environment along with the Pydicom 2.4.4 and NumPy 1.26.4 libraries. The DICOM file was read, and a new pixel matrix was created using the "Pixel Array" TAG. This matrix was adjusted with specific values from the file and normalized to ensure appropriate contrast. The resulting image was then saved in PNG format for further analysis and processing techniques application [20].

Advanced adaptive filtering techniques, for smoothing and for noise reduction, were combined with morphological operations to improve the quality of the tomographed PPE-Pb images, automatically cropping them to the region of interest (see Figure 1). Morphological opening and closing techniques with specific structuring kernels were used to remove noise components according to intensity and distribution variations in each image, as visual characteristics of PPE-Pb vary by manufacturer. This approach enabled progressive filtering without compromising the structural integrity of the cracks under analysis, preserving their essential morphological properties (see Figure 2).



Figure 1: Automated initial image segmentation process.



(a) Unprocessed image

(b) Noise-filtered image

Figure 2: Noise filtering with morphological opening and closing while preserving cracks.

2.2.2. Initial Crack Characterization

With PPE-Pb images in PNG format, a specialized algorithm was configured to detect cracks and pronounced folds. Adaptive filters were applied to smooth images when necessary, combined with iterative morphological techniques on grayscale versions. This process enhanced structures of interest, such as lead cracks and folds, while suppressing over 50% of noise elements unrelated to these features. Segmentation was performed using thresholding with a pixel intensity threshold of 255 to highlight crack and fold edges, facilitating contour delineation.

Crack detection involved two OpenCV-based processing techniques. The first applied morphological opening to remove minor imperfections, followed by brightness adjustment to improve structural visibility. A binary thresholding was used to generate a mask, and contours were detected using a hierarchical approach, focusing on outer contours only. Figure 3 illustrates this process. The second technique began with truncation thresholding to limit pixel intensity, followed by dilation and erosion operations to emphasize linear features. Canny edge detection was then applied to enhance crack definitions, and contours were extracted using the same hierarchical method. The refined contour segmentation from this strategy is shown in Figure 4, ensuring effective detection under various image conditions.



Figure 3: Brightness adjustment and thresholding to highlight features.



Figure 4: Crack detection with truncation thresholding and Canny-enhanced edges.

2.2.3. Preliminary Crack Selection.

An adaptive threshold-based segmentation method was used to generate an initial mask that includes potential cracks, folds, and external contours. The detected contours were classified according to their morphology and relative location. A proximity filtering algorithm was implemented to exclude contours not considered as cracks. For this purpose, the minimum distances between contours were calculated, and elements within a defined threshold distance were discarded. Finally, the selected cracks were projected onto the original image, and additional corrections were made when necessary to eliminate possible false positives. The final result was labeled for further analysis and quantification, as shown in Figure 5.

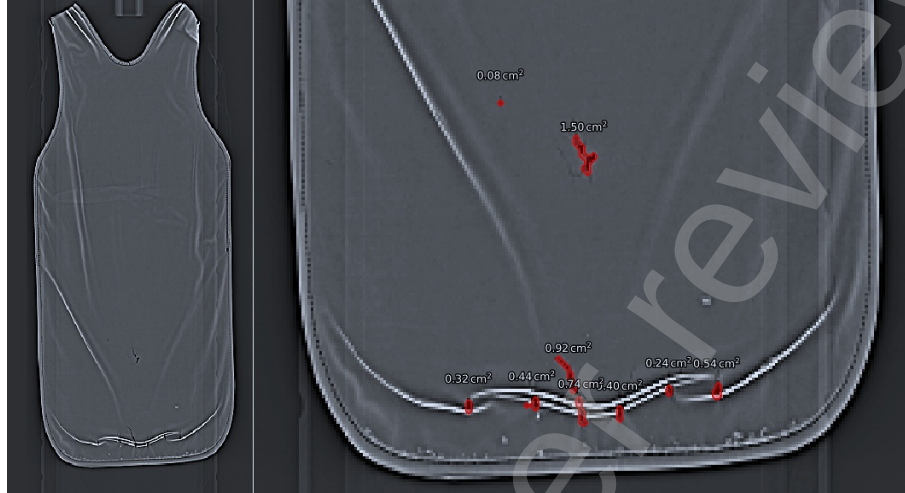


Figure 5: Crack detection.

2.3. Rejection and Acceptance Criteria

According to Phillips and Duran [1], equations have been developed to determine the minimum effective area that a Pb-based personal protective equipment (PPE-Pb) must maintain to ensure adequate protection against ionizing radiation. These equations establish objective criteria that integrate economic considerations and radiological safety.

The annual incremental cost associated with using PPE-Pb, denoted as I_c , is calculated as follows:

$$I_c = \frac{P_{PPE-Pb}}{T_L} \quad (1)$$

Where:

- P_{PPE-Pb} : Total Price of the PPE-Pb, including shipping costs and taxes.
- T_L : Service life of the PPE-Pb, as specified by the manufacturer.

From I_c , the annual incremental dose avoided (I_d) can be estimated as a function of the cost per millisievert (C_p) of radiation avoided per occupationally exposed worker:

$$I_d = \frac{I_c}{C_p} = \frac{P_{PPE-Pb}}{T_L C_p} \quad (2)$$

Where:

- C_p : Cost to prevent a worker's exposure to 1 mSv of radiation.

The percentage increase in dose due to the use of PPE-Pb, considering the presence of cracks or defects, is determined as follows:

$$\frac{I_d}{D_m} \cdot 100\% \quad (3)$$

Where:

- D_m : Maximum average annual dose received by a worker.

Additionally, the additional dose (d_i) caused by cracks in the PPE-Pb is described in Phillips and Duran [1] by the following equation:

$$d_i = \frac{I_d}{D_m} D_u T = \sum_t w_t D_u \frac{H_t}{A_t} - w_t D_u \frac{H_t}{A_t} T \quad (4)$$

Where:

- d_i : Additional dose attributed to cracks in the PPE-Pb.
- D_u : Unattenuated dose through the PPE-Pb.
- T : Transmission factor of the PPE-Pb.
- w_t : Tissue weighting factor for the region of interest.
- H_t : Total area of the cracks in the region of interest.
- A_t : Surface area of the region of interest.

Substituting equation (2) into equation (4), we obtain:

$$\frac{P_{PPE-Pb}}{T_L C_p D_m} T = \sum_t w_t \frac{H_t}{A_t} - w_t \frac{H_t}{A_t} T \quad (5)$$

Based on equation 5, specific rejection criteria can be established for different types of cracks in PPE-Pb. These criteria are detailed below.

2.3.1. Cracks Uniformly Distributed Across the Apron

In this case,

- $w_t = 1$
- $A_t = A$: Total surface area of the apron
- $H_t = H$

Replacing and solving for H in equation (5), we obtain the equation to determine the minimum crack area required to decide, based on regulatory parameters, whether the PPE-Pb should be replaced. This criterion applies to cracks uniformly distributed across the apron.

$$H = A \frac{P_{PPE-Pb}}{T_L C_p D_m} \frac{T}{1 - T} \quad (6)$$

2.3.2. Cracks Localized Exclusively in the Reproductive Region of the Apron

In this case, the weighted summation of equation (5) is given by:

$$\frac{P_{PPE-Pb}}{T_L C_p D_m} T = \left(w_g \frac{H_g}{A_g} - w_g T \frac{H_g}{A_g} \right) + \left(w_r \frac{H_r}{A_r} - w_r T \frac{H_r}{A_r} \right) \quad (7)$$

Where,

- w_g : Tissue weighting factor for the gonads $w_g = 0.2$ [2]
- w_r : Tissue weighting factor for other regions.
- A_g : Total surface area of the gonadal region $A_g = 20\text{cm}^2$ [21]
- A_r : Total surface area of other regions
- H_g : Area of cracks in the gonadal region.
- H_r : Area of cracks in the remaining regions.

Since only the reproductive region is of interest, we set $H_r = 0$:

$$\frac{P_{PPE-Pb}}{T_L C_p D_m} T = w_g \frac{H_g}{A_g} - w_g T \frac{H_g}{A_g} \quad (8)$$

Solving for H_g , we obtain the equation for determining the minimum crack area in the reproductive region required to decide, based on regulatory parameters, whether the PPE-Pb should be replaced.

$$H_g = A_g \frac{P_{PPE-Pb}}{T_L C_p D_m w_g} \frac{T}{1 - T} \quad (9)$$

2.3.3. Cracks in a One-Piece Thyroid Shield

For a single-piece thyroid protector, it can be assumed, according to Phillips and Duran [1], that the relative contribution of the equivalent dose received by the thyroid gland to the effective dose is determined by the weighting factors established in ICRP [2]. This is expressed by the following relation:

$$E = \sum w_t E_t \quad (10)$$

Where E represents the total effective dose and E_t the equivalent dose in tissue t. Considering the effective dose E as the maximum average annual dose received by a worker, denoted as D_m , and adopting the weighting factor for the thyroid as $w_t = w_{Th} = 0.05$ according to ICRP [2], it follows that the equivalent dose in the thyroid is:

$$E_t = \frac{D_m}{w_{Th}} \quad (11)$$

From this relation, the percentage increase in dose due to the presence of cracks or defects in the thyroid protector is expressed as:

$$\frac{I_d}{D_m} \cdot 100\% = \frac{I_d w_{Th}}{D_m} \cdot 100\% \quad (12)$$

Following a procedure analogous to that used in equation (4), the additional dose due to the presence of cracks in the thyroid protector is expressed as:

$$d_i = \frac{I_d w_{Th}}{D_m} D_u T = \sum_t w_t D_u \frac{H_t}{A_t} - w_t D_u \frac{H_t}{A_t} T \quad (13)$$

Particularizing for the case of the thyroid, we obtain:

$$\frac{I_d w_{Th}}{D_m} D_u T = w_{Th} D_u \frac{H_{Th}}{A_{Th}} - w_{Th} D_u \frac{H_{Th}}{A_{Th}} T \quad (14)$$

Solving for H_{Th} in equation (14), we obtain the expression that allows calculating the minimum area of the cracks in the thyroid region to determine, based on regulatory criteria, whether the thyroid protector should be replaced:

$$H_{Th} = A_{Th} \frac{P_{PPE} - P_{b_{Th}}}{T_L C_{p_{Th}} D_m} \frac{T}{1 - T} \quad (15)$$

3. Uncertainties

The evaluation of Type A uncertainty was carried out following Evaluation of measurement data — Guide to the expression of uncertainty in measurement [22]. The experimental protocol consisted of:

- Independent positioning of the same lead apron on 5 occasions (removing and repositioning the equipment for each test).
- Two experimental series differing only in the vertical position of the stretcher.
- Acquisition of 5 topograms per series.
- Automatic quantification of fissures using the software.

For each test series, the standard uncertainty was calculated as:

$$U_A = \frac{s}{\sqrt{n}} \quad (16)$$

Where s is the sample standard deviation of the detected fissure areas and $n = 5$ (topograms per series). The relative uncertainty was expressed as:

$$U_{\%} = \frac{U_A}{\bar{X}} \times 100\% \quad (17)$$

Table 1. Uncertainty in fissure area quantification (analysis software).

Series	Mean (cm ²)	Std. Dev. (cm ²)	U_A (cm ²)	$U_{\%}$ (%)
1	6.996	0.354	0.158	2.26
2	6.856	0.544	0.243	3.55
Combined	6.926	0.439	0.196	2.83

The results show that the software provides consistent measurements, with variability under 4% between repositionings. The slightly higher uncertainty in the upper position (3.55% vs 2.26%) could be related to geometric projection effects in the topogram or minor variations in apron coupling. The combined uncertainty of 2.83% suggests that:

- The detection algorithm is robust against positioning variations.
- The system's repeatability is suitable for clinical use.
- The vertical position of the stretcher does not introduce significant systematic errors.

4. Results

4.1 Classification of PPE-Pb and application of the rejection criterion through the algorithm

This section presents the results of the differentiation and quantification of PPE-Pb using the developed software. Unlike other approaches, the operation of this software is based on an automatic acceptance or rejection criterion, determined by a predefined threshold for the crack area. If cracks are detected with an area exceeding the established threshold (rejection criterion), the PPE-Pb is classified as "rejected". Conversely, if no cracks are detected or if they are below the threshold, the element is classified as "passed".

Once this classification is performed, the software generates a report with all relevant information, including the values of detected cracks and their automated evaluation. This report is sent to the health and safety personnel of the hospital or relevant service, thus completing the quality control process. This approach enables greater traceability, reduces subjectivity in evaluation, and improves inspection process efficiency, as discussed in similar works [23], [24], [25].

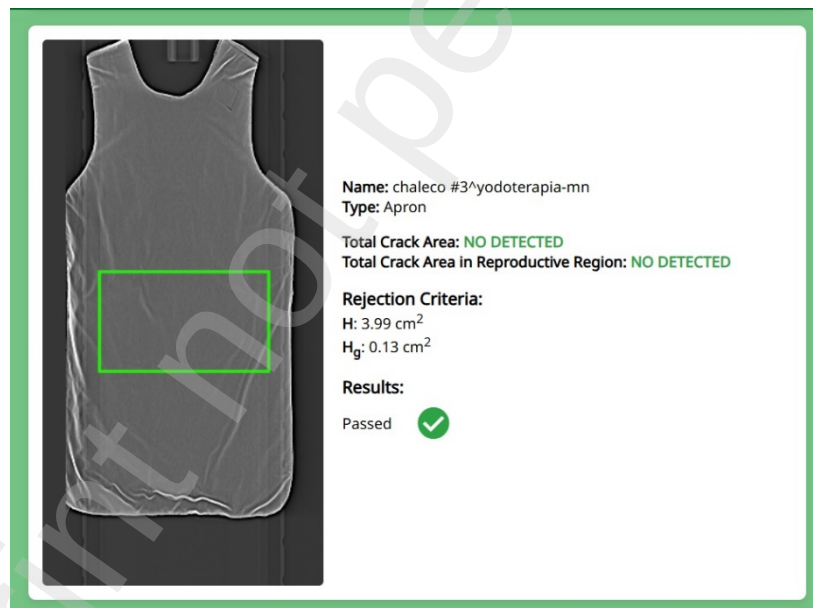


Figure 6: Inspection result: Vest in optimal condition, meets the required acceptance criteria to continue in service.

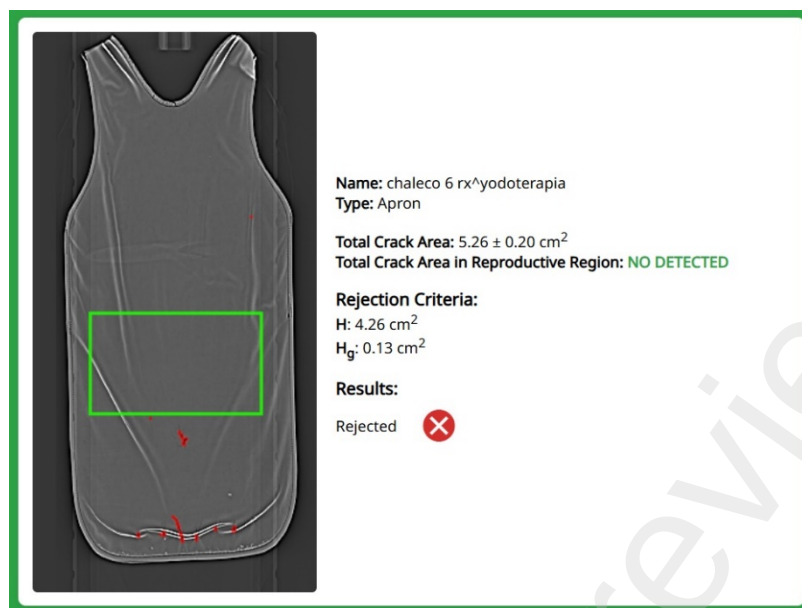


Figure 7: Inspection result: Vest in deficient condition, does not meet the required acceptance criteria to continue in service.

5. Discussion

The algorithm developed in this study successfully classified the available PPE-Pb, detecting and accurately separating elements with cracks and pronounced folds for subsequent analysis. This result highlights the effectiveness of the proposed method for classifying vests and thyroid protectors, as well as for detecting PPE-Pb with potential cracks in the lead. This high accuracy, coupled with correct identification of cracks and folds, provides a solid foundation for future implementations in automated PPE-Pb evaluation systems. However, the literature shows that there is still a lack of standardization in the methods and criteria used for the evaluation of these items, such as crack lengths, verifications of equivalent lead thickness, and visual inspections [23], [26], [27]. This study overcomes these limitations by introducing the quantification of crack areas, which has a direct impact on the radiobiological and economic evaluation of PPE-Pb, as proposed by previous models [1].

In terms of occupational health, the method used has significant implications, as the precise quantification of cracks contributes to a more accurate estimation of the effective dose received by workers exposed to radiation [1]. Furthermore, the developed software facilitates the analysis of PPE-Pb under radiobiological and economic criteria, making it a useful tool for decision-making in health institutions, thereby improving efficiency in quality control processes.

The need for periodic checks (PC) and follow-up of EPP-Pb, with a minimum annual frequency, has been widely documented in the international literature [23], [28]–[32]. In this context, the presented tool not only offers precise and reliable analysis but also significantly reduces the time spent on specialized work.

Despite the effectiveness of the method, it is important to recognize that due to the lack of clear regulations on the acceptance or rejection of PPE-Pb, each medical facility establishes its own criteria, leading to variability in evaluation procedures and the frequency of checks [7], [23], [26], [27], [29]–[33]. The approach of this study, based on the model by Phillips and Duran [1], allows for the objective quantification of crack areas, overcoming the limitations of traditional visual inspections and providing a more solid foundation for decision-making.

The methodology employed in this study also has the potential to be applied to the detection of cracks in other radioprotective items, such as containers and lead shields, thus broadening its field of application [9]. Compared to more complex detection techniques, such as deep learning and convolutional neural networks (CNN), the image processing techniques proposed in this work have the advantage of being simpler and faster, allowing for more direct manipulation of detection parameters [19], [34]. However, it is crucial to note that, prior to the application of computational tools, stricter acceptance and rejection criteria could be

adopted, based on methodological advancements and recent studies, which may include cost/benefit approaches. The adoption of these methodologies will depend on the decisions made by the radiological safety authorities at each institution.

6. Conclusions and future perspectives

In this work, a quality control approach for fissured lead aprons was developed using conventional image processing techniques, including spatial filters, segmentation, and labeling. This study represents a contribution toward the development of tools that facilitate radiological protection tasks in hospital settings and promote more efficient decision-making between occupational safety departments and those responsible for radiological protection in healthcare institutions. The developed algorithm is considered viable for ensuring the proper condition of lead PPE through periodic quality control.

Various authors emphasize the need, from a radiological protection perspective, for the implementation of quality control procedures on lead PPE to protect occupationally exposed personnel. However, there is a documented gap between the need for such controls and the existence of international documents or recommendations that guide the most efficient methodologies for their implementation, as well as the thresholds to be applied for fissure quantification and decision-making regarding the replacement of lead PPE in healthcare institutions [26], [35], [36].

The methodology applied in this study also allows for the identification of folds in lead PPE, which, according to the literature, require monitoring due to the potential development of fissures as a result of material wear.

The use of computational techniques for process optimization in the field of radiological protection, such as the one presented in this study, reduces the time required for this task in hospital environments—from five working days for manual analysis (using RadiAnt) to two days with this algorithm.

Finally, an algorithm has been presented that enables an automatic analysis of the quality control process for lead PPE, using acceptance or rejection criteria that involve both radiobiological and economic aspects. It also incorporates the quantification model published by other authors [1]. This tool can guide occupational health and safety and medical physics departments regarding the replacement of these protective elements, as needed.

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